Current Clarity Stage 13: Low-Visibility Engine

Diagnostic only.

- Production is unchanged, and nothing was deployed.
- No Sector field observation was used, the P2 fresh holdout was not touched, and Water Temperature was not started.
- Michael's Guntersville survey stays an acceptance test.

Freeze. The pre-registration was hashed at **2026-09-30T20:39:59Z** (sha256 `d44874fb…`), before any Stage 13 feature or metric was computed. Certification addendum 2 was frozen with it; the only later addendum, CERT-3 (21:07Z), sets the SYKE extraction budget (section 11).

Stage 12 erratum. Stage 12's clear-water line had reported the in-situ "clear" water type (268 rows), not Secchi ≥ 6 ft (574 rows). A key collision in `models12.py` caused it.

- **Corrected:** C3 on ACOLITE ×1.56 and –3.3 ft, against production ×1.97 and –5.2 ft.
- **The Stage 12 hand-off was reissued** with the correction. Nothing else in Stage 12 changes.

The answer

Limited water is largely fixed. Stained water is not, and it looks like a resolution limit rather than a missing signal.

- **The multicomponent models.** Red, red-edge, green/red, blue/green and brightness together cut the exact-feet error from ×1.47 (C3) to ×1.36–1.38.
- **Limited water.** Recall rises from 0.32 to 0.54–0.65, and every CI excludes 0.
- **Stained water (2–3 ft)** stays at 0.21–0.38 recall on every model. The two diagnostics say why:
  - the class is only a factor of 1.5 wide, narrower than the best retrieval's error spread;
  - an unbiased retrieval with that spread would recall stained only 33%;
  - even perfect in-situ turbidity and chlorophyll reach only 48%.
- **Low visibility is mostly particles.** In stained and limited water, in-situ turbidity explains 28–30% of the variance (most of what in-situ data explain). Dissolved organic carbon adds 8%, and chlorophyll about 0. Sentinel-2's blue/green colour feature adds nothing there.
- **No pre-registered candidate is a Stage 13 success.** All fail the stained bars (recall ≥ 0.45, precision ≥ 0.40).
- **BEST, by the frozen fallback rule, is O2:** the linear multicomponent model with class cut-points learned on held-out lakes. Its balanced accuracy is 0.568 against 0.501 (CI excludes 0).
  - It generalises across regions (0.92–0.93 of its lake-held-out stained and limited recall).
  - Its class never flips on identical S2A/S2C 60 m disks once ±12% boundary bands are used.
  - **Its cost:** normal recall falls from 0.64 to 0.41.
- **Coverage reaches 100%** in three labelled tiers:
  - 76% ACOLITE default;
  - 12% relaxed mask (as good as tier 1);
  - 12% Sen2Cor fallback (lower authority; poor in murky water).
- **The blind set is still not ready.** Expanding SYKE fills the total and the cool season, but not the US stained, sediment and limited counts.

1. What was done

Foundation (unchanged)	ACOLITE K4, lake-elevation pressure, S2C V-ρw, and C3 on ACOLITE as benchmark A (identical to Stage 12: maximum difference 0.0 ft). Whole-lake folds; Guntersville excluded from every fit
Rows	**1,206 station-days at 792 lakes** with a valid ACOLITE disk (E13); coverage judged on all 1,582 processed rows
Mechanism	Same-day in-situ turbidity and chlorophyll (509 rows), plus colour and dissolved/total organic carbon fetched for corpus sites only (true colour 131 rows, DOC 190, TOC 212)
Features	Seven, pre-registered: red, NDCI, blue/green, green/red, 740/665, brightness, and QAA Kd(492)
Candidates	A; the QAA-switch Secchi model (pure and calibrated); C-lin; C-gbm (constrained boosting); O1 ordinal logit; O2 cut-points; and the Dogliotti red/NIR switch
Coverage	Relaxed ACOLITE masks rerun on the 313 scenes with a lost station; Sen2Cor-C3 as a lower-authority fallback
Footprint	The 12 tandem pairs rerun keeping the grids; 60 m disks on a 300 m grid
Certification	SYKE expanded from 400 to all 3,705 eligible lakes, value-blind

2. Deviations and notes

- **CERT-3 (the extraction budget).** It covers all 687 prescreened cool-season SYKE events and 1,500 seeded warm ones. Readiness could not change either way, because the US core carries the per-class check.
- **The QAA-switch clamps a negative spectral slope η to 0** rather than refusing. The pre-registration did not say.
- **O2's feet** are C-lin's continuous prediction (O2 is a class model). The Stage 12 exact-feet rule is applied to O2's predicted class.
- **The footprint Δft rule** uses the Stage 12 definition: the median over pairs of each pair's median. The mean over pairs is also shown, in the file.
- **"Not materially worse" (coverage)** is judged two ways, and both pass. One compares the union with tier 1 counting Unknown as a miss, as registered. The other compares it with tier 1's own rows, which is stricter.
- **No absorbance-at-440 nm results** exist at the corpus sites. The CDOM mechanism uses true colour and DOC/TOC instead.
- **A flaw in one pre-registered rule, disclosed below.** The attenuation verdict tests recall only.

3. What limits stained and limited water

Held-out R^2 of log Secchi, split by component (Shapley):

Held-out R^2 of log Secchi, Shapley shares	Components
In-situ turbidity + chlorophyll, all rows (n 509)	particles (turbidity) 0.50; pigment (chlorophyll) 0.17 (total 0.67)
… stained + limited	particles (turbidity) 0.30; pigment (chlorophyll) 0.01 (total 0.31)
With DOC/TOC, all rows (n 155, 112 lakes)	particles (turbidity) 0.31; pigment (chlorophyll) 0.20; dissolved colour (DOC/TOC) 0.21 (total 0.72)
… stained + limited (n 64)	particles (turbidity) 0.28; pigment (chlorophyll) 0.00; dissolved colour (DOC/TOC) 0.08 (total 0.36)
With true colour, all rows (n 69, 49 lakes)	particles (turbidity) 0.23; pigment (chlorophyll) 0.11; dissolved colour (true colour) 0.29 (total 0.63)
… stained + limited (n 38)	particles (turbidity) 0.01; pigment (chlorophyll) -0.02; dissolved colour (true colour) 0.01 (total 0.00)
Satellite feature groups (C-lin), all rows	particles 0.29; dissolved colour 0.13; visibility ratio (green/red) 0.19; pigment 0.04 (total 0.64)
… stained + limited	particles 0.19; dissolved colour 0.00; visibility ratio (green/red) 0.09; pigment 0.04 (total 0.32)

How much class recall is even reachable:

Recall (precision)	Clear	Normal	Stained	Limited
A perfect in-situ retrieval: turbidity + chlorophyll known, held out	0.68 (0.82)	0.62 (0.49)	0.48 (0.42)	0.62 (0.82)
An unbiased retrieval with C-gbm's error spread	0.86	0.51	0.33	0.76
Class width (log10)	open	0.301	0.176	open

- **Particles dominate low visibility.**
  - In stained and limited water, turbidity explains most of what in-situ data can explain: 0.30 of 0.31 held-out R^2 (stained and limited rows among the 509 matched), and 0.28 of 0.36 where DOC exists.
  - Dissolved carbon adds 0.08. True colour, on only 38 stained or limited rows, adds nothing measurable.
  - Across all water, dissolved colour does matter (0.21–0.29 of R^2). That is mainly what separates clear lakes from coloured ones.
- **The satellite sees the particles, not the colour, in murky water.** Within stained and limited rows, the particle features carry 0.19 and green/red 0.09. The blue/green colour feature carries 0.00.
- **Stained water is a resolution problem.**
  - **The class is narrow.** 2–3 ft is a factor of 1.5 (0.176 in log10).
  - **The retrieval is wider.** The best held-out retrieval (C-gbm) has an interquartile error of −0.14 to +0.12.
  - **The arithmetic.** Blur the true values with that error, and even an unbiased estimator recalls stained only 33%, normal 51%, limited 76% and clear 86%.
  - **Even perfect knowledge falls short.** Known in-situ turbidity and chlorophyll recall stained only 48%.
  - **So** no satellite feature can be expected to lift stained recall past about 0.4–0.5 while the classes stay this narrow.
- **Limited water was a model-form problem.** C3's single red-band form flattens in bright, murky water. The multicomponent features (740/665, green/red, brightness) keep the gradient, and limited recall doubles.

4. The candidates

Candidate (held out by whole lakes, 1,206 rows at 792 lakes)	Answers	Median |log| (factor)	Clear R / P	Normal R / P	Stained R / P	Limited R / P	Balanced accuracy	Ordinal κ
A: C3 on ACOLITE (benchmark)	100%	0.169 (×1.47)	0.75 / 0.80	0.64 / 0.41	**0.29 / 0.32**	**0.32 / 0.81**	0.501	0.66
B-pure: QAA-switch Secchi, no fit	93%	0.258 (×1.81)	0.51 / 0.82	0.38 / 0.43	**0.11 / 0.54**	**0.71 / 0.26**	0.428	0.36
B-cal: QAA-switch, calibrated	93%	0.252 (×1.79)	0.60 / 0.68	0.56 / 0.28	**0.03 / 0.09**	**0.00 / 0.00**	0.295	0.27
C-lin: 6-feature linear	100%	0.140 (×1.38)	0.86 / 0.84	0.58 / 0.47	**0.25 / 0.35**	**0.54 / 0.68**	0.561	0.75
C-gbm: constrained boosting (7 features)	100%	0.134 (×1.36)	0.82 / 0.87	0.64 / 0.47	**0.24 / 0.32**	**0.57 / 0.70**	0.566	0.76
O1: ordinal logit	100%	class only	0.88 / 0.81	0.51 / 0.48	**0.21 / 0.33**	**0.64 / 0.65**	0.561	0.77
O2: C-lin with learned class cut-points	100%	class only	0.84 / 0.83	0.41 / 0.46	**0.38 / 0.36**	**0.65 / 0.57**	0.568	0.76
Champion form on red turbidity	100%	0.189 (×1.54)	0.69 / 0.77	0.65 / 0.35	**0.20 / 0.30**	**0.23 / 0.76**	0.439	0.56
Champion form on red/NIR-switched turbidity	100%	0.189 (×1.54)	0.69 / 0.77	0.65 / 0.35	**0.20 / 0.30**	**0.21 / 0.76**	0.437	0.55

R / P is recall / precision; κ is quadratic-weighted.

Gains against A:

Against A (lake bootstrap 95% CI)	Stained recall	Limited recall	Balanced accuracy
B-pure	-0.18 (-0.26 to -0.11)	+0.38 (+0.30 to +0.47)	-0.073 (-0.11 to -0.04)
C-lin	-0.04 (-0.12 to +0.04)	+0.22 (+0.13 to +0.31)	+0.060 (+0.03 to +0.10)
C-gbm	-0.06 (-0.14 to +0.02)	+0.24 (+0.16 to +0.32)	+0.065 (+0.03 to +0.10)
O1	-0.08 (-0.17 to +0.01)	+0.32 (+0.24 to +0.40)	+0.060 (+0.03 to +0.09)
O2	+0.08 (-0.03 to +0.19)	+0.32 (+0.24 to +0.41)	+0.067 (+0.03 to +0.10)

The pre-registered success criteria:

Candidate	1. Stained ≥ 0.45 / 0.40	2. Limited ≥ 0.50 / 0.60	3. Both gains' CI > 0	4. Clear and normal kept
B-pure	**fail**	**fail**	**fail**	**fail**
B-cal	**fail**	**fail**	**fail**	**fail**
C-lin	**fail**	pass	**fail**	**fail**
O1	**fail**	pass	**fail**	**fail**
O2	**fail**	**fail**	**fail**	**fail**
C-gbm	**fail**	pass	**fail**	pass

Region-held-out (success criterion 5, for BEST and A):

Region-held-out (5 US regions)	Clear	Normal	Stained	Limited	Stained ratio	Limited ratio
A	0.73	0.61	0.27	0.24	0.92	0.73
O2	0.79	0.52	0.35	0.60	0.92	0.93

- **Limited water: fixed by the multicomponent models.**
  - Recall is 0.54 (C-lin), 0.57 (C-gbm), 0.64 (O1) and 0.65 (O2), against 0.32 for A. Every gain's CI excludes 0.
  - Precision is 0.57–0.70.
- **Stained water: not fixed.** No model's stained recall gain has a CI above 0. O2 gets closest, at +0.08 (–0.03 to +0.19).
- **Feet: the best in the series.** C-gbm reaches ×1.36 and C-lin ×1.38, against ×1.47 for A. Ordinal κ rises from 0.66 to 0.75–0.77.
- **No candidate is a SUCCESS.**
  - Criterion 1 (stained recall ≥ 0.45) fails for all.
  - C-gbm passes 2 and 4, and fails 1 and 3.
  - O2 fails all four. Its stained gain's CI includes 0 (criteria 1 and 3), its limited precision is 0.57 against the 0.60 bar (criterion 2), and it trades normal recall for stained (criterion 4).
- **BEST is O2, by the frozen fallback rule:**
  - the highest balanced accuracy (0.568), beating A by +0.067 with the CI excluding 0;
  - it holds up region-held-out (stained ratio 0.92, limited 0.93);
  - C-gbm is statistically indistinguishable (0.566), keeps normal recall at 0.64, and gives the best feet.
- **The ordinal models do not beat regression-then-threshold.** Their class skill equals C-lin's, except where O2's learned cut-points move the 3 ft and 2 ft boundaries to 3.78 and 2.12 ft. That simply buys stained recall with normal recall.

5. The red/NIR turbidity switch

Red/NIR switch (Dogliotti 2015)	Value
Rows where the switch activates (ρ665 > 0.05)	2.4% (fully NIR 0.8%)
Stained, same-day turbidity (n 35)	red 0.235 → switch 0.264 (-12%; CI of the difference +0.000 to +0.054)
Limited, same-day turbidity (n 40)	red 0.317 → switch 0.317 (+0%; CI of the difference -0.102 to +0.068)
Sediment, same-day turbidity (n 18)	red 0.229 → switch 0.314 (-37%; CI of the difference +0.000 to +0.284)
Champion form, switched against red: recall change	stained +0.00, limited -0.01

RED/NIR TURBIDITY SWITCH USEFUL: NO.

- **It barely activates.** The red band exceeds the switch threshold (ρ665 > 0.05) at 2.4% of stations.
- **Where it does, it helps nowhere.** Stained error worsens by 12%, limited is unchanged, and sediment worsens by 37%. Stained and limited recall do not rise.
- **Red-band saturation is not what limits these waters** in this corpus. They are rarely bright enough.

6. The direct attenuation model

QAA-switch diagnostics	Value
Reference band used	560 nm 246, 665 nm 493, 740 nm 467 rows
Refusals	a<aw 82, range 1; answered 1123
B-pure: coverage; stained+limited recall gain (CI)	93%; +0.098 (+0.044 to +0.158)
B-pure: median |log| minus A's (CI)	+0.065 to +0.120

DIRECT ATTENUATION MODEL USEFUL: YES by the frozen rule, but not in substance.

- **The rule was flawed:** it tested stained + limited recall and nothing else.
- **B-pure passes it by calling murky water limited far too often.** Limited recall is 0.71, but precision is 0.26.
- **On every other measure it is worse than A:**
  - stained recall 0.11;
  - balanced accuracy 0.43 against 0.50 (CI excludes 0);
  - exact-feet error ×1.81 against ×1.47.
- **Calibrating it (B-cal) does not help.** Its Secchi estimate barely correlates with the truth once refitted.
- **Why the physics fails.** The inversion took the 740 nm reference on 39% of rows. There, assuming a(740) ≈ a_w inflates backscatter and attenuation in turbid inland water.
- **The QAA Kd(492) is still a useful input,** as one of C-gbm's seven features.

7. The dissolved-colour (CDOM) signal

Removing F_bg (blue/green) from	Stained recall change (CI)	Balanced accuracy change (CI)
C-lin	+0.052 (-0.013 to +0.123)	-0.006 (-0.031 to +0.021)
O1	+0.035 (-0.007 to +0.079)	+0.015 (-0.001 to +0.031)

CDOM/ABSORPTION SIGNAL USEFUL: NO.

- **Removing the blue/green feature lowers nothing.** Stained recall even rises slightly (+0.05 in C-lin, +0.04 in O1), and neither change is significant.
- **The feature carries information only between clear and coloured lakes** (0.13 of R^2 over all rows), not inside stained or limited water (0.00).
- **The in-situ data agree:** dissolved carbon is a minor factor in low visibility, next to particles.

8. Coverage

Relaxed-mask variant (tier 2)	Stations recovered	Their median |log| (A)	Tier-1 median |log| (A)	Passes
M1	32	0.189	0.169	yes
M2	158	0.158	0.169	yes
M3	191	0.173	0.169	yes

Coverage tiers (all 1,582 processed rows; recall/precision, Unknown = miss)	Value
O2: coverage	tier 1 76% · tier 2 12% · tier 3 12% · unknown 0%
O2, Tier 1 only	balanced 0.432; clear 0.64/0.83; normal 0.31/0.47; stained 0.29/0.36; limited 0.49/0.57
O2, Tiers 1–2	balanced 0.507; clear 0.71/0.83; normal 0.37/0.48; stained 0.37/0.37; limited 0.58/0.55
O2, Tiers 1–3 (tier 3 lower authority)	balanced 0.547; clear 0.80/0.83; normal 0.42/0.44; stained 0.37/0.34; limited 0.60/0.55
O2, tier 2 rows only (n 191)	balanced 0.603; clear 0.80/0.75; normal 0.35/0.59; stained 0.47/0.44; limited 0.79/0.47
O2, tier 3 rows only (n 185)	balanced 0.359; clear 0.60/0.89; normal 0.60/0.25; stained 0.12/0.07; limited 0.11/0.43
A: coverage	tier 1 76% · tier 2 12% · tier 3 12% · unknown 0%
A, Tier 1 only	balanced 0.381; clear 0.58/0.80; normal 0.48/0.41; stained 0.23/0.32; limited 0.24/0.81
A, Tiers 1–2	balanced 0.440; clear 0.64/0.79; normal 0.58/0.42; stained 0.27/0.32; limited 0.27/0.81
A, Tiers 1–3 (tier 3 lower authority)	balanced 0.480; clear 0.73/0.80; normal 0.62/0.40; stained 0.28/0.28; limited 0.28/0.77
A, tier 2 rows only (n 191)	balanced 0.474; clear 0.83/0.69; normal 0.58/0.49; stained 0.26/0.30; limited 0.21/0.75
A, tier 3 rows only (n 185)	balanced 0.359; clear 0.60/0.89; normal 0.60/0.25; stained 0.12/0.07; limited 0.11/0.43

The relaxed masks recover real water.

- **Variant M3** ignores the negative-reflectance flag (with positive visible medians) and the SWIR non-water flag where NDWI > 0.
- **What it recovers.** 191 of the 353 lost stations, whose error is the same as tier 1's (A: 0.173 against 0.169). Class skill on those rows is good: balanced accuracy 0.60, stained 0.47 recall.
- **The two flags in the default mask** that dropped good water are the SWIR non-water flag (158 stations) and the negative-reflectance flag (32).

Coverage recovered, with authority stated:

- tier 1: 76.2% (ACOLITE, default mask);
- tier 2: 12.1% (ACOLITE, relaxed mask; equal authority on this evidence);
- tier 3: 11.7% (C3 on Sen2Cor; **lower authority**);
- **total 100%,** or **88.3% at ACOLITE authority** (just under the 90% bar without the fallback).

Tier 3 is honest but weak. On its rows, stained recall is 0.13 and limited 0.11 (balanced 0.36). The fallback should show a class with a low-confidence marker, not exact feet. The union is not materially worse (balanced accuracy 0.547, against 0.568 on tier 1's own rows).

9. Class boundaries at the product footprint

Engine, 60 m disks on identical tandem water (10 pairs)	Raw S2C, hard	Harmonised, hard	Harmonised, with bands	Median |Δft|	Corpus predictions in a band
A	11.1%	8.2%	**1.0%**	0.10 ft	29%
C-lin	9.5%	4.0%	**0.0%**	0.13 ft	24%
C-gbm	12.0%	8.1%	**0.5%**	0.15 ft	23%
O2	10.0%	3.8%	**0.0%**	0.13 ft	25%

S2A/B/C PLATFORM-INVARIANT AT PRODUCT FOOTPRINT: YES (for BEST, with bands).

- **O2 never flips class** on identical S2A/S2C 60 m disks once the ±12% boundary bands are applied, over 10 scored pairs (Volta and Nasser have no valid disk). The median change is 0.13 ft.
- **Without bands it flips on 3.8%,** against C3's 8.2% (11% per pixel in Stage 12).
- **The cost.** A quarter of corpus predictions fall inside a band. Those would be shown as a boundary state, for example "normal-to-stained", not a single class.
- **What changed from Stage 12.** Stage 12's 11% flips were per pixel. At the real footprint, and with a model that uses several bands rather than C3's NDCI branch switch, the S2C difference no longer changes the class.

10. Capability (historical, lake-held-out; not readiness)

What a user would get (BEST, tiers 1–3)	Share
Exact feet (and the class)	95%
Class only	5%
Unknown	0%

Capability	Exact feet (published rows)	Feet verdict	Clarity class	Class verdict
Clear	721 published (97%); ×1.40; -2.3 ft	**PARTIAL**	recall 0.80, precision 0.83, Unknown 0%	**SUPPORTED**
Normal	344 published (90%); ×1.34; +0.4 ft	**SUPPORTED**	recall 0.42, precision 0.44, Unknown 0%	**NOT**
Stained	209 published (92%); ×1.39; +0.6 ft	**SUPPORTED**	recall 0.37, precision 0.34, Unknown 0%	**NOT**
Limited	224 published (98%); ×1.59; +0.7 ft	**PARTIAL**	recall 0.60, precision 0.55, Unknown 0%	**NOT**
Algae	222 published (92%); ×1.41; +0.7 ft	**SUPPORTED**	class accuracy 0.55 vs constant 0.37	**SUPPORTED**
Sediment	89 published (97%); ×1.28; +0.1 ft	**SUPPORTED**	bright sediment called clear 0% (n 25); accuracy 0.50 vs 0.42	**PARTIAL**
Bottom-risk	–	**NOT**	no depth input	**NOT**
Score		**5.0 / 7 → 7/10**		**2.5 / 7 → 4/10**

- **EXACT-FEET CAPABILITY: 7/10** (5.0 of 7), up from 0/10.
  - The multicomponent feet are good enough (×1.28–1.41 in every class but limited) that the inner folds authorise exact feet for almost every predicted class. Feet are shown for 95% of rows.

    - **SUPPORTED:** normal (+0.4 ft), stained (+0.6 ft), algae and sediment.
    - **PARTIAL:**
        - clear, a −2.3 ft median (the ±1.5 ft SUPPORTED bar is missed; deep clear water is still under-read);
        - limited, a median |log| of 0.201 against the 0.20 bar (×1.59).
- **CLARITY-CLASS CAPABILITY: 4/10** (2.5 of 7), up from 1/10.
    - **SUPPORTED:** clear (recall 0.80, precision 0.83) and algae (accuracy 0.55 against a constant 0.37).
    - **PARTIAL:** sediment.
    - **Not met:** normal, stained and limited miss the 0.60 recall/precision bars. O2 lowered normal recall to buy stained recall.
- **The two scores diverge because they measure different things.** Feet are judged on typical error, and classes on hitting a narrow band. That is the resolution finding again.

11. The blind certification set

Blind check (Stage 12 rule)	Value	Bar	
Matched observations	914 (NLA 165, SYKE 749)	≥ 400	pass
Distinct lakes	821	≥ 100	pass
US states (US core)	39	≥ 20	pass
Largest state share (US core)	9.7%	≤ 15%	pass
October–April observations	19.5% (163 lakes)	≥ 15% across ≥ 20 lakes	pass
Expected clear (US core)	77	≥ 40	pass
Expected normal (US core)	41	≥ 40	pass
Expected stained (US core)	24	≥ 60	**fail**
Expected algae (US core)	36	≥ 30	pass
Expected sediment (US core)	13	≥ 20	**fail**
Expected limited (US core)	23	≥ 25	**fail**

BLIND CERTIFICATION SET READY: NO. The expansion fixed five of the six Stage 12 checks. The sixth is structural.

- **What the full SYKE pool added.** The 3,705 eligible Finnish lakes yielded 2,187 extracted events and 625 new matches. The set grows from 289 to **914 observations at 821 lakes** (NLA 165, SYKE 749).
- **Cool-season water now passes:** 19.5% October–April across 163 lakes, against the 15% bar. Every cool-season row is Finnish, and most are October. The US core has none.
- **What still fails** is the expected class mix of the US core, the part that represents Sector's water:
    - stained 24 against a minimum of 60;
    - sediment 13 against 20;
    - limited 23 against 25.
- **SYKE cannot fill those counts.** Finnish lakes are humic, rarely turbid, and do not stand in for US reservoirs.
- **To close the gap needs new US data.** One route is a TVA or USACE records request for reservoir Secchi. The other is your approval to open the sealed WQP tier, which was downloaded in Stage 1 but never analysed.
- **The set was built value-blind.** It holds ids, dates and scene items only, and no Secchi value was read. `blind_sample13.csv` is hashed in the freeze file (sha256 `bcb18659…`, 21:59Z).

12. Guntersville (historical agency rows, reported apart)

Guntersville, 42 historical rows at 10 station groups (reported apart)	Error	Median signed	Class accuracy
A	×1.34 (0.128)	-0.7 ft	0.52
C-lin	×1.28 (0.108)	-0.5 ft	0.64
C-gbm	×1.43 (0.155)	-1.2 ft	0.45
O2	×1.28 (0.108)	-0.5 ft	0.62

- **C-lin and O2 improve Guntersville** (42 historical rows at 10 station groups): ×1.28 and −0.5 ft, against C3's ×1.34 and −0.7 ft. O2 recalls 4 of the 7 stained rows, against C3's 1.
- **C-gbm does worse there** (×1.43). It may fit national patterns that Guntersville's arms do not follow.
- **This is historical only.** Michael's survey remains the independent acceptance test.

13. What still stands between this and 9/10

Product readiness stays 2/10 (the Stage 10 rule: fresh evidence only).

What is left:

1. **The class definition limits stained recall.** 2–3 ft is narrower than any satellite retrieval can resolve, and narrower even than in-situ turbidity and chlorophyll can. There are three honest paths:
   - accept a boundary state ("normal-to-stained") as a supported answer;
   - widen or merge the stained class for Sector Intelligence;
   - require a much more precise retrieval, which nothing tested here provides.
2. **Normal recall (0.41–0.64) trades against stained recall.** One cut-point set cannot maximise both.
3. **Tier-3 coverage is weak.** The Sen2Cor fallback misclassifies murky water, so 12% of stations need either another ACOLITE recovery path or a class-only low-confidence answer.
4. **Clear water is still under-read by about 2–3 ft** in deep lakes. That blocks clear-feet SUPPORTED.
5. **No blind US certification set exists without new data.** US stained, sediment, limited and cool-season counts need either a TVA/USACE records request or your approval of a sealed, never-analysed WQP tier.
6. **Fresh evidence** (P2's fresh data and the certification) is still the only thing that moves readiness.
7. **Bottom-risk** needs a depth layer (the Navionics track).

Files

Everything is in `scratchpad/stage13/`:

Path	Contents
`PREREGISTRATION_STAGE13.md`, `FREEZE_STAGE13.json`, `cert/CERT_ADDENDUM_2.md`, `cert/CERT_ADDENDUM_3.md`	The frozen plan, its hashes and the addenda
`insitu/`	The colour/DOC fetch for corpus sites, and the same-day matches
`analysis/`	`common13.py` (features, QAA-switch, switch, metrics), `models13.py`, `mechanism13.py`, `coverage13.py`, `capscore13.py`, JSON results
`coverage/`	The relaxed-mask reruns (313 scenes)
`tandem/`	The footprint disks and `footprint13.json`
`cert/`	The SYKE pool, prescreen, extraction, `blind_sample13.csv` and its checks

PRIMARY CAUSE OF STAINED/LIMITED FAILURE: Resolution, not a missing signal. Stained water (2–3 ft) is a band only a factor of 1.5 wide, narrower than the best retrieval error (×1.36; interquartile −0.14 to +0.12 log), so even an unbiased estimator recalls it only ~33% (perfect in-situ turbidity and chlorophyll reach 48%). Low visibility is mostly particles (in stained and limited water, in-situ turbidity explains 28–30% of the variance, most of what in-situ data explain; DOC 8%; chlorophyll ~0). Limited water was failing because C3's single red-band form flattens in murky water, which the multicomponent features fix.

CDOM/ABSORPTION SIGNAL USEFUL: NO

RED/NIR TURBIDITY SWITCH USEFUL: NO

DIRECT ATTENUATION MODEL USEFUL: YES (by the frozen rule only: the QAA-switch model buys limited recall by over-calling limited water, precision 0.26, and is worse than C3 on balanced accuracy and feet)

BEST ENGINE: O2: the linear multicomponent model (red, NDCI, blue/green, green/red, 740/665, brightness) with class cut-points learned on held-out lakes (6.0 / 3.78 / 2.12 ft). It is BEST by the fallback rule, not a pre-registered SUCCESS (stained bars unmet). Balanced accuracy 0.568 against 0.501 for C3; C-gbm is equivalent and keeps normal recall.

CLEAR RECALL / PRECISION: 0.80 / 0.83 (O2 on all 1,582 rows, tiers 1–3; 0.84 / 0.83 on the 1,206 ACOLITE-valid rows; C3 0.73 / 0.80)

NORMAL RECALL / PRECISION: 0.42 / 0.44 (0.41 / 0.46 on ACOLITE-valid rows; C3 0.62 / 0.40). O2 buys stained recall with normal recall; C-gbm keeps normal at 0.64 / 0.47

STAINED RECALL / PRECISION: 0.37 / 0.34 (0.38 / 0.36 on ACOLITE-valid rows; C3 0.28 / 0.28). The gain is not significant, and it is below the 0.45 / 0.40 bar

LIMITED RECALL / PRECISION: 0.60 / 0.55 (0.65 / 0.57 on ACOLITE-valid rows; C3 0.28 / 0.77). Recall gain +0.32, CI excludes 0

COVERAGE: 100% in three tiers: 76.2% ACOLITE default mask + 12.1% ACOLITE relaxed mask (88.3% at ACOLITE authority) + 11.7% Sen2Cor-C3 fallback at lower authority

S2A/B/C PLATFORM-INVARIANT AT PRODUCT FOOTPRINT: YES

BLIND CERTIFICATION SET READY: NO

CLARITY-CLASS CAPABILITY: 4/10 (historical; clear and algae SUPPORTED, sediment PARTIAL; normal, stained and limited below the 0.60 bars)

EXACT-FEET CAPABILITY: 7/10 (historical; normal, stained, algae and sediment SUPPORTED; clear and limited PARTIAL; feet published for 95% of rows)

REMAINING BLOCKERS TO 9/10: (1) the stained class (2–3 ft) is narrower than any retrieval's error, and than in-situ turbidity can resolve: it needs a boundary-state answer ("normal-to-stained") or a wider class, not a new signal; (2) normal and stained recall trade off through one set of cut-points; (3) the Sen2Cor fallback (tier 3, 12% of stations) misreads murky water; (4) deep clear water is under-read by about 2.3 ft; (5) the blind US set lacks stained, sediment and limited water (expected 24/13/23 against 60/20/25), which needs TVA/USACE records or your approval of the sealed WQP tier; (6) fresh evidence: readiness stays 2/10 until P2 and certification run; (7) bottom-risk needs depth data